Toxidromes and general poisoning care

Toxidromes

- Anticholinergic: hot, dry, mydriasis, delirium, retention
- Cholinergic: SLUDGE, miosis, bradycardia, bronchorrhoea
- Sympathomimetic: hypertension, tachycardia, mydriasis, sweating
- Opioid: coma, miosis, respiratory depression; sedative-hypnotic: coma, normal pupils

General management

- ABC, glucose, ECG, temperature; history and pill counts
- Decontamination: activated charcoal within 1 h if airway safe
- Enhanced elimination: dialysis (lithium, salicylate, methanol), urinary alkalinisation
- Antidotes: naloxone, flumazenil (rare), NAC, fomepizole

Specific treatment

- Depends on toxin; consult poison centre

Key point: Recognise the toxidrome, then give the antidote.

Organophosphate poisoning

Mechanism

- Inhibits acetylcholinesterase → muscarinic + nicotinic + CNS effects
- Muscarinic: SLUDGE, bronchorrhoea, bradycardia; nicotinic: fasciculation, weakness

Management

- Decontaminate, **atropine** (double dose every 5 min until chest dry, HR > 80, SBP > 80)
- Pralidoxime (controversial), benzodiazepines for seizures; intubate (avoid suxamethonium)

Intermediate syndrome

- 24–96 h: proximal weakness, cranial nerves, respiratory failure
- Delayed polyneuropathy (2–3 weeks)

Key point: Titrate atropine to a dry chest, not to pupils.

Guidelines (no 2026 note on this topic)

Aluminium phosphide poisoning

Toxicity

- Phosphine gas: inhibits cytochrome oxidase → cellular hypoxia
- Refractory shock, myocardial toxicity, metabolic acidosis, ARDS

Management

- No antidote; supportive; avoid water lavage (releases gas)
- **Magnesium**, calcium, bicarbonate, vasopressors, IABP/ECMO

Poor prognostic factors

- Shock, pH < 7.2, ECG changes, large dose, delay

Key point: Supportive care only: early ECMO may help.

Tricyclic antidepressant poisoning

Mechanism

- Na-channel block (wide QRS), alpha-block (hypotension), anticholinergic, GABA block (seizures)

Management

- **Sodium bicarbonate 1–2 mEq/kg** if QRS > 100 ms or arrhythmia; target pH 7.45–7.55
- Benzodiazepines for seizures; intralipid; avoid physostigmine and flumazenil

Monitoring

- ECG for ≥ 6 h; QRS normal and asymptomatic → discharge

Key point: Wide QRS in TCA overdose: give bicarbonate.

Salicylate poisoning

Acid–base

- Early respiratory alkalosis (respiratory centre stimulation), then HAGMA (uncoupling)

Management

- **Urinary alkalinisation** (bicarbonate, urine pH 7.5–8.5); correct K
- Avoid intubation (worsens acidosis); if needed, maintain hyperventilation

Haemodialysis indications

- Level > 7.2 mmol/L (100 mg/dL), CNS symptoms, AKI, pulmonary oedema, refractory acidosis

Key point: Don't let intubation stop the hyperventilation.

Calcium channel blocker and beta-blocker overdose

Comparison

- CCB: hyperglycaemia, vasodilation (amlodipine)
- Beta-blocker: hypoglycaemia, bradycardia, bronchospasm

Stepwise management

- Fluids, calcium, atropine, vasopressors
- **High-dose insulin euglycaemia** (1 U/kg bolus, 1–10 U/kg/h)
- Glucagon (beta-blocker), lipid emulsion, pacing, ECMO

Monitoring HDI

- Glucose every 30 min, K, echo

Key point: High-dose insulin is the key therapy.

Guidelines (no 2026 note on this topic)

Carbon monoxide and cyanide poisoning

CO pathophysiology

- Binds Hb (240× O2) → left shift; mitochondrial toxicity

Management

- **100% O2** (half-life 4–6 h → 74–90 min)
- **Hyperbaric O2**: COHb > 25%, LOC, neuro signs, pregnancy, ischaemia

Cyanide

- Fire smoke, lactate > 8 mmol/L, high venous O2
- **Hydroxocobalamin 5 g**

Key point: Smoke victim with high lactate → give hydroxocobalamin.

Serotonin syndrome, NMS and malignant hyperthermia

Comparison

- Serotonin: rapid onset, clonus, hyperreflexia, diarrhoea
- NMS: slow onset, rigidity, bradyreflexia; MH: anaesthetic triggers, hypercarbia

Serotonin management

- Stop drugs, benzodiazepines, cooling, **cyproheptadine**

NMS and MH

- NMS: stop drug, dantrolene, bromocriptine
- MH: stop volatiles, **dantrolene 2.5 mg/kg**, cooling

Key point: Clonus = serotonin; rigidity = NMS.